

GENERAL APTITUDE

Previous Year Questions • 2010–2024

GATE — ALL BRANCHES

Himanshu Agarwal & Anish Saha

PrepFusion™

Where Concept Meets Clarity!

Preface

Welcome to PrepFusion™!

This book is a complete compilation of the questions GATE has actually asked in General Aptitude (all GATE branches), gathered from the original papers and arranged year by year and session by session. Nothing is paraphrased or reconstructed: every question appears as it was set, tagged with its year, session and marks.

Each question carries an identifier that maps one-to-one onto its worked solution in the companion solutions volume, so you can move between practice and explanation without a lookup table. Answer keys close every section, and answers revised by the examining authority (MTA) are marked where they occur.

Used end to end, the book shows you what the paper rewards — which topics repeat, how the framing has shifted over the years, and where your preparation still has gaps.

Meet your Authors

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> *"We built this book the way every aspirant deserves one to be built — organized strictly by year and by session, so that every hour of practice maps directly onto how GATE actually tests you."*
>
> — Himanshu Agarwal & Anish Saha

How to Read the Question Numbers

Every question in this book carries a three-part identifier instead of a plain serial number:

Q24.5.3

24	GATE exam year — 2024 (two digits; 98 = 1998)
5	Session number of that year's exam
3	Question number within that session

So **Q24.5.3** is the 3rd aptitude question of Session 5, GATE 2024. Numbers are therefore not sequential through the book — they tell you the year and session at a glance. The same identifier is used in the questions book, the solutions book and the answer keys, so you can jump between them without a lookup table. In the PDF bookmarks panel, expand a session to see every question ID and click one to go straight to it.

What the Bubbles Mean

Every question ends with three empty circles for you to shade in yourself:

  Easy  Medium  Hard

The companion solutions volume marks the same question with *our* recommended difficulty, using the identical bubbles. If you think a rating should be different, tell us directly — report it under the **Study Hub** section at **pyq.prepfusion.in**.

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Scan any code with your phone camera, or tap the link beneath it. **pyq.prepfusion.in** is also where you can read this book's worked solutions online — and in its **Study Hub** section, report an error in any question or request a video solution for it.

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YEAR

2014

9 Sessions · 90 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AE, AR, CE, CH, CY, EY, MA, MT, PH, XE	10
Session 2	AG, BT, CE, GG, IN, MN, PI, TF, XL	10
Session 3	EC, ME	10
Session 4	EC, ME	10
Session 5	EC, ME	10
Session 6	EC, ME	10
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Year Snapshot

Total Questions: 90 **MCQ:** 71 **MSQ:** 0 **NAT:** 19

Session : 1 | Forenoon | 02/02/14

Q14.1.1 MCQ 2014 1M

A student is required to demonstrate a high level of comprehension of the subject, especially in the social sciences.
The word closest in meaning to comprehension is

A. understanding **B.** meaning

C. concentration **D.** stability

Q14.1.2 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
One of his biggest _____ was his ability to forgive.

A. vice **B.** virtues **C.** choices **D.** strength

Q14.1.3 MCQ 2014 1M

Rajan was not happy that Sajan decided to do the project on his own. On observing his unhappiness, Sajan explained to Rajan that he preferred to work independently.
Which one of the statements below is logically valid and can be inferred from the above sentences?

A. Rajan has decided to work only in a group.

B. Rajan and Sajan were formed into a group against their wishes.

C. Sajan had decided to give in to Rajan's request to work with him.

D. Rajan had believed that Sajan and he would be working together.

Q14.1.4 MCQ 2014 1M

If $y = 5x^2 + 3$, then the tangent at $x = 0, y = 3$

A. passes through $x = 0, y = 0$ **B.** has a slope of $+1$

C. is parallel to the x-axis **D.** has a slope of -1

Q14.1.5 NAT 2014 1M

A foundry has a fixed daily cost of Rs $50,000$ whenever it operates and a variable cost of Rs $800Q$, where Q is the daily production in tonnes. What is the cost of production in Rs per tonne for a daily production of 100 tonnes?

Q14.1.6 MCQ 2014 1M

Find the odd one in the following group: ALRVX, EPVZB, ITZDF, OYEIK

A. ALRVX **B.** EPVZB **C.** ITZDF **D.** OYEIK

Q14.1.7 MCQ 2014 2M

Anuj, Bhola, Chandan, Dilip, Eswar and Faisal live on different floors in a six-storeyed building (the ground floor is numbered 1, the floor above it 2, and so on). Anuj lives on an even-numbered floor. Bhola does not live on an odd numbered floor. Chandan does not live on any of the floors below Faisal's floor. Dilip does not live on floor number 2. Eswar does not live on a floor immediately above or immediately below Bhola. Faisal lives three floors above Dilip. Which of the following floor-person combinations is correct?

	Anuj	Bhola	Chandan	Dilip	Eswar	Faisal
(A)	6	2	5	1	3	4
(B)	2	6	5	1	3	4
(C)	4	2	6	3	1	5
(D)	2	4	6	1	3	5

Q14.1.8 NAT 2014 2M

The smallest angle of a triangle is equal to two thirds of the smallest angle of a quadrilateral. The ratio between the angles of the quadrilateral is 3 : 4 : 5 : 6. The largest angle of the triangle is twice its smallest angle. What is the sum, in degrees, of the second largest angle of the triangle and the largest angle of the quadrilateral?

Q14.1.9 MCQ 2014 2M

One percent of the people of country X are taller than 6 ft. Two percent of the people of country Y are taller than 6 ft. There are thrice as many people in country X as in country Y. Taking both countries together, what is the percentage of people taller than 6 ft?

A. 3.0 **B.** 2.5 **C.** 1.5 **D.** 1.25

Q14.1.10 MCQ 2014 2M

The monthly rainfall chart based on 50 years of rainfall in Agra is shown in the following figure. Which of the following are true? (k percentile is the value such that k percent of the data fall below that value)

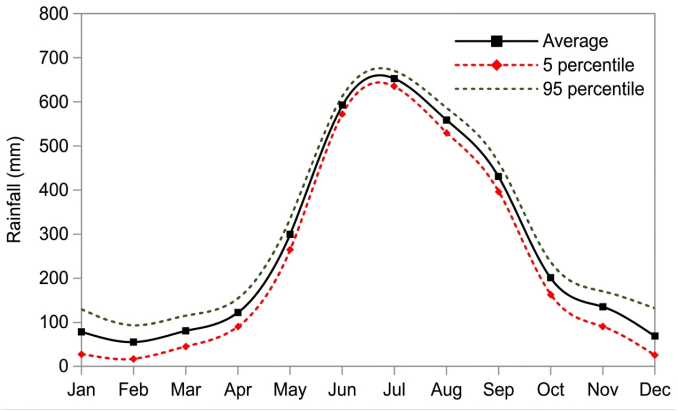


Fig. Q14.1.10

(i) On average, it rains more in July than in December
(ii) Every year, the amount of rainfall in August is more than that in January
(iii) July rainfall can be estimated with better confidence than February rainfall
(iv) In August, there is at least 500 mm of rainfall

A. (i) and (ii) **B.** (i) and (iii)

C. (ii) and (iii) **D.** (iii) and (iv)

Session : 2 | Afternoon | 02/02/14

Q14.2.1 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
A person suffering from Alzheimer's disease _____ short-term memory loss.

A. experienced **B.** has experienced

C. is experiencing **D.** experiences

Q14.2.2 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
_____ is the key to their happiness; they are satisfied with what they have.

A. Contentment **B.** Ambition

C. Perseverance **D.** Hunger

Q14.2.3 MCQ 2014 1M

Which of the following options is the closest in meaning to the sentence below?
"As a woman, I have no country."

A. Women have no country.

B. Women are not citizens of any country.

C. Women's solidarity knows no national boundaries.

D. Women of all countries have equal legal rights.

Q14.2.4 NAT 2014 1M

In any given year, the probability of an earthquake greater than Magnitude 6 occurring in the Garhwal Himalayas is 0.04. The average time between successive occurrences of such earthquakes is _____ years.

Q14.2.5 MCQ 2014 1M

The population of a new city is 5 million and is growing at 20% annually. How many years would it take to double at this growth rate?

A. 3-4 years **B.** 4-5 years **C.** 5-6 years **D.** 6-7 years

Q14.2.6 MCQ 2014 2M

In a group of four children, Som is younger to Riaz. Shiv is elder to Ansu. Ansu is youngest in the group. Which of the following statements is/are required to find the eldest child in the group?

Statements

1. Shiv is younger to Riaz.
2. Shiv is elder to Som.

A. Statement 1 by itself determines the eldest child.

B. Statement 2 by itself determines the eldest child.

C. Statements 1 and 2 are both required to determine the eldest child.

D. Statements 1 and 2 are not sufficient to determine the eldest child.

Q14.2.7 MCQ 2014 2M

Moving into a world of big data will require us to change our thinking about the merits of exactitude. To apply the conventional mindset of measurement to the digital, connected world of the twenty-first century is to miss a crucial point. As mentioned earlier, the obsession with exactness is an artefact of the information-deprived analog era.

When data was sparse, every data point was critical, and thus great care was taken to avoid letting any point bias the analysis.
From *"BIG DATA" Viktor Mayer-Schonberger and Kenneth Cukier*
The main point of the paragraph is:

A. The twenty-first century is a digital world

B. Big data is obsessed with exactness

C. Exactitude is not critical in dealing with big data

D. Sparse data leads to a bias in the analysis

Q14.2.8 MCQ | 2014 | 2M

The total exports and revenues from the exports of a country are given in the two pie charts below. The pie chart for exports shows the quantity of each item as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 5 lakh tonnes and the total revenues are 250 crore rupees. What is the ratio of the revenue generated through export of Item 1 per kilogram to the revenue generated through export of Item 4 per kilogram?

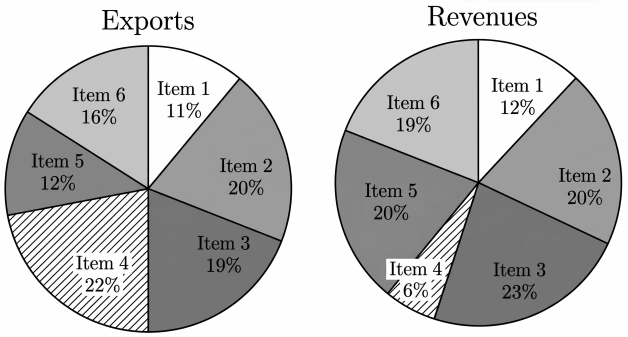


Fig. Q14.2.8

A. 1:2 **B.** 2:1 **C.** 1:4 **D.** 4:1

Q14.2.9 MCQ | 2014 | 2M

X is 1 km northeast of Y. Y is 1 km southeast of Z. W is 1 km west of Z. P is 1 km south of W. Q is 1 km east of P. What is the distance between X and Q in km?

A. 1 **B.** $\sqrt{2}$ **C.** $\sqrt{3}$ **D.** 2

Q14.2.10 NAT | 2014 | 2M

10% of the population in a town is HIV^+. A new diagnostic kit for HIV detection is available; this kit correctly identifies HIV^+ individuals 95% of the time, and HIV^- individuals 89% of the time. A particular patient is tested using this kit and is found to be positive. The probability that the individual is actually positive is _____

Session : 3 | Forenoon | 15/02/14

Q14.3.1 MCQ | 2014 | 1M

Choose the most appropriate phrase from the options given below to complete the following sentence
The aircraft _____ take off as soon as its flight plan was filed.

A. is allowed to **B.** will be allowed to

C. was allowed to **D.** has been allowed to

Q14.3.2 MCQ | 2014 | 1M

Read the statements:

All women are entrepreneurs.
Some women are doctors.

Which of the following conclusions can be logically inferred from the above statements?

A. All women are doctors

B. All doctors are entrepreneurs

C. All entrepreneurs are women

D. Some entrepreneurs are doctors

Q14.3.3 MCQ | 2014 | 1M

Choose the most appropriate word from the options given below to complete the following sentence.
Many ancient cultures attributed disease to supernatural causes. However, modern science has largely helped _____ such notions.

A. impel **B.** dispel **C.** propel **D.** repel

Q14.3.4 MCQ | 2014 | 1M

The statistics of runs scored in a series by four batsmen are provided in the following table. Who is the most consistent batsman of these four?

Batsman	Average	Standard deviation
K	31.2	5.21
L	46.0	6.35
M	54.4	6.22
N	17.9	5.90

A. K **B.** L **C.** M **D.** N

Q14.3.5 NAT 2014 1M

What is the next number in the series?

12 35 81 173 357 _____

Q14.3.6 MCQ 2014 2M

Find the odd one from the following group:

W,E,K,O I,Q,W,A F,N,T,X N,V,B,D

A. W,E,K,O **B.** I,Q,W,A **C.** F,N,T,X **D.** N,V,B,D

Q14.3.7 MCQ 2014 2M

For submitting tax returns, all resident males with annual income below Rs 10 lakh should fill up Form P and all resident females with income below Rs 8 lakh should fill up Form Q. All people with incomes above Rs 10 lakh should fill up Form R, except non residents with income above Rs 15 lakhs, who should fill up Form S. All others should fill Form T. An example of a person who should fill Form T is

A. a resident male with annual income Rs 9 lakh

B. a resident female with annual income Rs 9 lakh

C. a non-resident male with annual income Rs 16 lakh

D. a non-resident female with annual income Rs 16 lakh

Q14.3.8 NAT 2014 2M

A train that is 280 metres long, travelling at a uniform speed, crosses a platform in 60 seconds and passes a man standing on the platform in 20 seconds. What is the length of the platform in metres?

Q14.3.9 MCQ 2014 2M

The exports and imports (in crores of Rs.) of a country from 2000 to 2007 are given in the following bar chart. If the trade deficit is defined as excess of imports over exports, in which year is the trade deficit 1/5th of the exports?

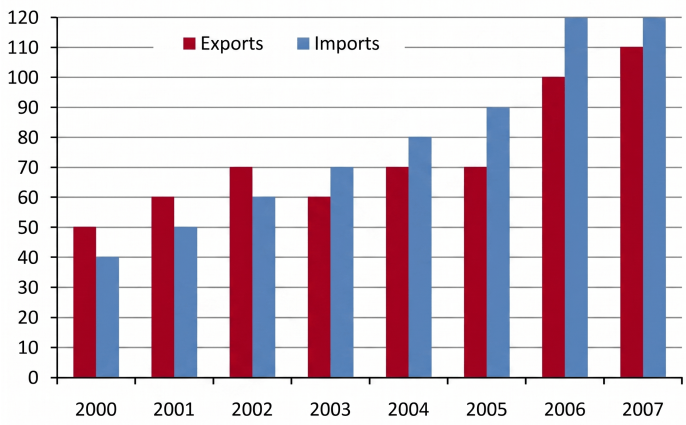


Fig. Q14.3.9

A. 2005 **B.** 2004 **C.** 2007 **D.** 2006

Q14.3.10 MCQ 2014 2M

You are given three coins: one has heads on both faces, the second has tails on both faces, and the third has a head on one face and a tail on the other. You choose a coin at random and toss it, and it comes up heads. The probability that the other face is tails is

A. 1/4 **B.** 1/3 **C.** 1/2 **D.** 2/3

Session : 4 | Afternoon | 15/02/14

Q14.4.1 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.

Communication and interpersonal skills are _____ important in their own ways.

A. each **B.** both **C.** all **D.** either

Q14.4.2 MCQ 2014 1M

Which of the options given below best completes the following sentence?

She will feel much better if she _____.

A. will get some rest **B.** gets some rest

C. will be getting some rest **D.** is getting some rest

Q14.4.3 MCQ 2014 1M

Choose the most appropriate pair of words from the options given below to complete the following sentence.
She could not _____ the thought of _____ the election to her bitter rival.

A. bear, loosing **B.** bare, loosing

C. bear, losing **D.** bare, losing

Q14.4.4 MCQ 2014 1M

A regular die has six sides with numbers 1 to 6 marked on its sides. If a very large number of throws show the following frequencies of occurrence: 1 → 0.167; 2 → 0.167; 3 → 0.152; 4 → 0.166; 5 → 0.168; 6 → 0.180. We call this die

A. irregular **B.** biased **C.** Gaussian **D.** insufficient

Q14.4.5 NAT 2014 1M

Fill in the missing number in the series.
2 3 6 15 _____ 157.5 630

Q14.4.6 MCQ 2014 2M

Find the odd one in the following group

Q,W,Z,B B,H,K,M W,C,G,J M,S,V,X

A. Q,W,Z,B **B.** B,H,K,M **C.** W,C,G,J **D.** M,S,V,X

Q14.4.7 MCQ 2014 2M

Lights of four colors (red, blue, green, yellow) are hung on a ladder. On every step of the ladder there are two lights. If one of the lights is red, the other light on that step will always be blue. If one of the lights on a step is green, the other light on that step will always be yellow. Which of the following statements is not necessarily correct?

A. The number of red lights is equal to the number of blue lights

B. The number of green lights is equal to the number of yellow lights

C. The sum of the red and green lights is equal to the sum of the yellow and blue lights

D. The sum of the red and blue lights is equal to the sum of the green and yellow lights

Q14.4.8 NAT 2014 2M

The sum of eight consecutive odd numbers is 656. The average of four consecutive even numbers is 87. What is the sum of the smallest odd number and second largest even number?

Q14.4.9 MCQ 2014 2M

The total exports and revenues from the exports of a country are given in the two charts shown below. The pie chart for exports shows the quantity of each item exported as a percentage of the total quantity of exports. The pie chart for the revenues shows the percentage of the total revenue generated through export of each item. The total quantity of exports of all the items is 500 thousand tonnes and the total revenues are 250 crore rupees. Which item among the following has generated the maximum revenue per kg?

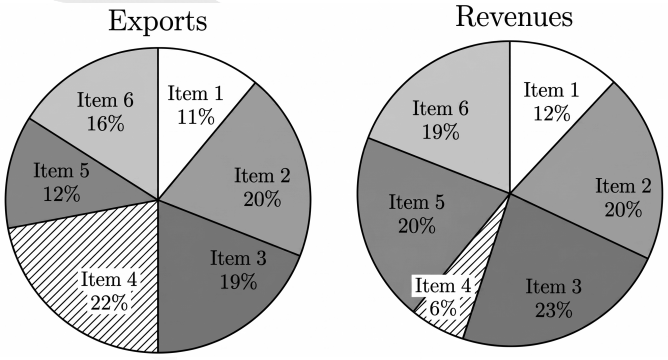


Fig. Q14.4.9

A. Item 2 **B.** Item 3 **C.** Item 6 **D.** Item 5

Q14.4.10 MCQ 2014 2M

It takes 30 minutes to empty a half-full tank by draining it at a constant rate. It is decided to simultaneously pump water into the half-full tank while draining it. What is the rate at which water has to be pumped in so that it gets fully filled in 10 minutes?

A. 4 times the draining rate

B. 3 times the draining rate

C. 2.5 times the draining rate

D. 2 times the draining rate

Session : 5 | Forenoon | 16/02/14

Q14.5.1 MCQ 2014 1M

"India is a country of rich heritage and cultural diversity." Which one of the following facts best supports the claim made in the above sentence?

A. India is a union of 28 states and 7 union territories.

B. India has a population of over 1.1 billion.

C. India is home to 22 official languages and thousands of dialects.

D. The Indian cricket team draws players from over ten states.

Q14.5.2 MCQ 2014 1M

The value of one U.S. dollar is 65 Indian Rupees today, compared to 60 last year. The Indian Rupee has _____.

A. depressed **B.** depreciated

C. appreciated **D.** stabilized

Q14.5.3 MCQ 2014 1M

'Advice' is _____.

A. a verb **B.** a noun

C. an adjective **D.** both a verb and a noun

Q14.5.4 NAT 2014 1M

The next term in the series 81, 54, 36, 24, _____ is

Q14.5.5 MCQ 2014 1M

In which of the following options will the expression $P < M$ be definitely true?

A. $M < R > P > S$ **B.** $M > S < P < F$

C. $Q < M < F = P$ **D.** $P = A < R < M$

Q14.5.6 MCQ 2014 2M

Find the next term in the sequence: 7G, 11K, 13M, ___

A. 15Q **B.** 17Q **C.** 15P **D.** 17P

Q14.5.7 MCQ 2014 2M

The multi-level hierarchical pie chart shows the population of animals in a reserve forest. The correct conclusions from this information are:



Fig. Q14.5.7

(i) Butterflies are birds
(ii) There are more tigers in this forest than red ants
(iii) All reptiles in this forest are either snakes or crocodiles
(iv) Elephants are the largest mammals in this forest

A. (i) and (ii) only **B.** (i), (ii), (iii) and (iv)

C. (i), (iii) and (iv) only **D.** (i), (ii) and (iii) only

Q14.5.8 NAT 2014 2M

A man can row at 8 km per hour in still water. If it takes him thrice as long to row upstream, as to row downstream, then find the stream velocity in km per hour.

Q14.5.9 NAT 2014 2M

A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. If the company registered a profit of Rs. 10 lakhs in 2012, at what price (in Rs.) was each air purifier sold?

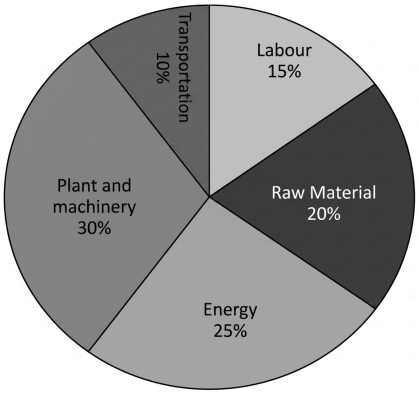


Fig. Q14.5.9

Q14.5.10 NAT 2014 2M

A batch of one hundred bulbs is inspected by testing four randomly chosen bulbs. The batch is rejected if even one of the bulbs is defective. A batch typically has five defective bulbs. The probability that the current batch is accepted is

Session : 6 | Afternoon | 16/02/14

Q14.6.1 MCQ 2014 1M

Which of the following options is the closest in meaning to the word underlined in the sentence below?
In a democracy, everybody has the freedom to disagree with the government.

A. dissent **B.** descent **C.** decent **D.** decadent

Q14.6.2 MCQ 2014 1M

After the discussion, Tom said to me, 'Please revert!'. He expects me to _____.

A. retract **B.** get back to him

C. move in reverse **D.** retreat

Q14.6.3 MCQ 2014 1M

While receiving the award, the scientist said, "I feel vindicated". Which of the following is closest in meaning to the word "vindicated"?

A. punished **B.** substantiated

C. appreciated **D.** chastened

Q14.6.4 MCQ 2014 1M

Let $f(x, y) = x^n y^m = P$. If x is doubled and y is halved, the new value of f is

A. $2^{n-m}P$ **B.** $2^{m-n}P$ **C.** $2(n-m)P$ **D.** $2(m-n)P$

Q14.6.5 NAT 2014 1M

In a sequence of 12 consecutive odd numbers, the sum of the first 5 numbers is 425. What is the sum of the last 5 numbers in the sequence?

Q14.6.6 MCQ 2014 2M

Find the next term in the sequence: 13M, 17Q, 19S, ___

A. 21W **B.** 21V **C.** 23W **D.** 23V

Q14.6.7 MCQ 2014 2M

If 'KCLFTSB' stands for 'best of luck' and 'SHSWDG' stands for 'good wishes', which of the following indicates 'ace the exam'?

A. MCHTX **B.** MXHTC

C. XMHCT **D.** XMHTC

Q14.6.8 MCQ 2014 2M

Industrial consumption of power doubled from 2000-2001 to 2010-2011. Find the annual rate of increase in percent assuming it to be uniform over the years.

A. 5.6 **B.** 7.2 **C.** 10.0 **D.** 12.2

Q14.6.9 NAT 2014 2M

A firm producing air purifiers sold 200 units in 2012. The following pie chart presents the share of raw material, labour, energy, plant & machinery, and transportation costs in the total manufacturing cost of the firm in 2012. The expenditure on labour in 2012 is Rs. 4,50,000. In 2013, the raw material expenses increased by 30% and all other expenses increased by 20%. What is the percentage increase in total cost for the company in 2013?

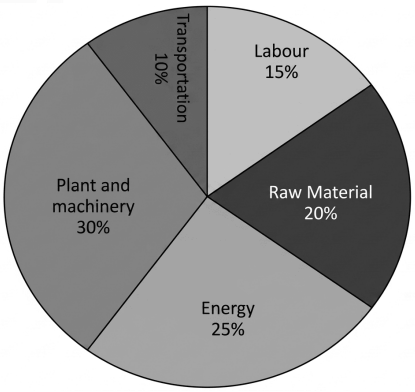


Fig. Q14.6.9

Q14.6.10 MCQ 2014 2M

A five digit number is formed using the digits 1,3,5,7 and 9 without repeating any of them. What is the sum of all such possible five digit numbers?

A. 6666660 **B.** 6666600 **C.** 6666666 **D.** 6666606

Session : 7 | Afternoon | 01/03/14

Q14.7.1 MCQ 2014 1M

Which of the following options is the closest in meaning to the phrase underlined in the sentence below?
It is fascinating to see life forms cope with varied environmental conditions.

A. adopt to **B.** adapt to **C.** adept in **D.** accept with

Q14.7.2 MCQ 2014 1M

Choose the most appropriate word from the options given below to complete the following sentence.
He could not understand the judges awarding her the first prize, because he thought that her performance was quite _____.

A. superb **B.** medium **C.** mediocre **D.** exhilarating

Q14.7.3 MCQ 2014 1M

In a press meet on the recent scam, the minister said, "The buck stops here". What did the minister convey by the statement?

A. He wants all the money

B. He will return the money

C. He will assume final responsibility

D. He will resist all enquiries

Q14.7.4 NAT 2014 1M

If $(z + 1/z)^2 = 98$, compute $(z^2 + 1/z^2)$.

Q14.7.5 MCQ 2014 1M

The roots of $ax^2 + bx + c = 0$ are real and positive. a, b and c are real. Then $ax^2 + b|x| + c = 0$ has

A. no roots **B.** 2 real roots

C. 3 real roots **D.** 4 real roots

Q14.7.6 MCQ 2014 2M

The Palghat Gap (or Palakkad Gap), a region about 30 km wide in the southern part of the Western Ghats in India, is lower than the hilly terrain to its north and south. The exact reasons for the formation of this gap are not clear. It results in the neighbouring regions of Tamil Nadu getting more rainfall from the South West monsoon and the neighbouring regions of Kerala having higher summer temperatures.
What can be inferred from this passage?

A. The Palghat gap is caused by high rainfall and high temperatures in southern Tamil Nadu and Kerala

B. The regions in Tamil Nadu and Kerala that are near the Palghat Gap are low-lying

C. The low terrain of the Palghat Gap has a significant impact on weather patterns in neighbouring parts of Tamil Nadu and Kerala

D. Higher summer temperatures result in higher rainfall near the Palghat Gap area

Q14.7.7 MCQ 2014 2M

Geneticists say that they are very close to confirming the genetic roots of psychiatric illnesses such as depression and schizophrenia, and consequently, that doctors will be able to eradicate these diseases through early identification and gene therapy.
On which of the following assumptions does the statement above rely?

A. Strategies are now available for eliminating psychiatric illnesses

B. Certain psychiatric illnesses have a genetic basis

C. All human diseases can be traced back to genes and how they are expressed

D. In the future, genetics will become the only relevant field for identifying psychiatric illnesses

Q14.7.8 NAT 2014 2M

Round-trip tickets to a tourist destination are eligible for a discount of 10% on the total fare. In addition, groups of 4 or more get a discount of 5% on the total fare. If the one way single person fare is Rs 100, a group of 5 tourists purchasing round-trip tickets will be charged Rs _____.

Q14.7.9 NAT 2014 2M

In a survey, 300 respondents were asked whether they own a vehicle or not. If yes, they were further asked to mention whether they own a car or scooter or both. Their responses are tabulated below. What percent of respondents do not own a scooter?

		Men	Women
Own vehicle	Car	40	34
	Scooter	30	20
	Both	60	46
Do not own vehicle		20	50

Q14.7.10 NAT 2014 2M

When a point inside of a tetrahedron (a solid with four triangular surfaces) is connected by straight lines to its corners, how many (new) internal planes are created with these lines? _____

Session : 8 | Forenoon | 02/03/14

Q14.8.1 MCQ 2014 1M

Choose the most appropriate phrase from the options given below to complete the following sentence.
India is a post-colonial country because

A. it was a former British colony

B. Indian Information Technology professionals have colonized the world

C. India does not follow any colonial practices

D. India has helped other countries gain freedom

Q14.8.2 MCQ 2014 1M

Who _____ was coming to see us this evening?

A. you said **B.** did you say

C. did you say that **D.** had you said

Q14.8.3 MCQ 2014 1M

Match the columns.

Column 1	Column 2
1) eradicate	P) misrepresent
2) distort	Q) soak completely
3) saturate	R) use
4) utilize	S) destroy utterly

A. 1:S, 2:P, 3:Q, 4:R **B.** 1:P, 2:Q, 3:R, 4:S

C. 1:Q, 2:R, 3:S, 4:P **D.** 1:S, 2:P, 3:R, 4:Q

Q14.8.4 MCQ 2014 1M

What is the average of all multiples of 10 from 2 to 198?

A. 90 **B.** 100 **C.** 110 **D.** 120

Q14.8.5 MCQ 2014 1M

The value of $\sqrt{12+\sqrt{12+\sqrt{12+\ldots}}}$ is

A. 3.464 **B.** 3.932 **C.** 4.000 **D.** 4.444

Q14.8.6 MCQ 2014 2M

The old city of Koenigsberg, which had a German majority population before World War 2, is now called Kaliningrad. After the events of the war, Kaliningrad is now a Russian territory and has a predominantly Russian population. It is bordered by the Baltic Sea on the north and the countries of Poland to the south and west and Lithuania to the east respectively. Which of the statements below can be inferred from this passage?

A. Kaliningrad was historically Russian in its ethnic make up

B. Kaliningrad is a part of Russia despite it not being contiguous with the rest of Russia

C. Koenigsberg was renamed Kaliningrad, as that was its original Russian name

D. Poland and Lithuania are on the route from Kaliningrad to the rest of Russia

Q14.8.7 MCQ 2014 2M

The number of people diagnosed with dengue fever (contracted from the bite of a mosquito) in north India is twice the number diagnosed last year. Municipal authorities have concluded that measures to control the mosquito population have failed in this region.
Which one of the following statements, if true, does not contradict this conclusion?

A. A high proportion of the affected population has returned from neighbouring countries where dengue is prevalent

B. More cases of dengue are now reported because of an increase in the Municipal Office's administrative efficiency

C. Many more cases of dengue are being diagnosed this year since the introduction of a new and effective diagnostic test

D. The number of people with malarial fever (also contracted from mosquito bites) has increased this year

Q14.8.8 MCQ 2014 2M

If x is real and $|x^2 - 2x + 3| = 11$, then possible values of $|-x^3 + x^2 - x|$ include

A. 2, 4 **B.** 2, 14 **C.** 4, 52 **D.** 14, 52

Q14.8.9 NAT 2014 2M

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students doubled in 2009, by what percent did the number of male students increase in 2009?

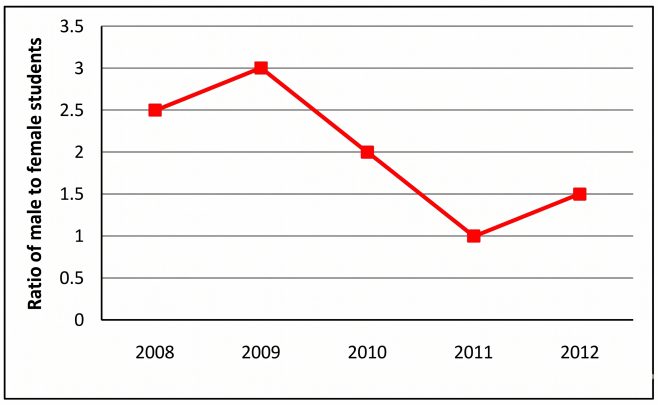


Fig. Q14.8.9

Q14.8.10 MCQ 2014 2M

At what time between 6 a.m. and 7 a.m. will the minute hand and hour hand of a clock make an angle closest to 60°?

A. 6:22 a.m. **B.** 6:27 a.m. **C.** 6:38 a.m. **D.** 6:45 a.m.

Session : 9 | Afternoon | 02/03/14

Q14.9.1 MCQ 2014 1M

While trying to collect (I) an envelope from under the table, (II) Mr. X fell down (III) and was losing consciousness. (IV)

Which one of the above underlined parts of the sentence is NOT appropriate?

A. I **B.** II **C.** III **D.** IV

Q14.9.2 MCQ 2014 1M

If she _____ how to calibrate the instrument, she _____ done the experiment.

A. knows, will have

B. knew, had

C. had known, could have

D. should have known, would have

Q14.9.3 MCQ 2014 1M

Choose the word that is opposite in meaning to the word "coherent".

A. sticky **B.** well-connected

C. rambling **D.** friendly

Q14.9.4 MCQ 2014 1M

Which number does not belong in the series below?
2, 5, 10, 17, 26, 37, 50, 64

A. 17 **B.** 37 **C.** 64 **D.** 26

Q14.9.5 MCQ 2014 1M

The table below has question-wise data on the performance of students in an examination. The marks for each question are also listed. There is no negative or partial marking in the examination.

Q No.	Marks	Answered Correctly	Answered Wrongly	Not Attempted
1	2	21	17	6
2	3	15	27	2
3	2	23	18	3

What is the average of the marks obtained by the class in the examination?

A. 1.34 **B.** 1.74 **C.** 3.02 **D.** 3.91

Q14.9.6 MCQ 2014 2M

A dance programme is scheduled for 10.00 a.m. Some students are participating in the programme and they need to come an hour earlier than the start of the event. These students should be accompanied by a parent. Other students and parents should come in time for the programme. The instruction you think that is appropriate for this is

A. Students should come at 9.00 a.m. and parents should come at 10.00 a.m.

B. Participating students should come at 9.00 a.m. accompanied by a parent, and other parents and students should come by 10.00 a.m.

C. Students who are not participating should come by 10.00 a.m. and they should not bring their parents. Participating students should come at 9.00 a.m.

D. Participating students should come before 9.00 a.m. Parents who accompany them should come at 9.00 a.m. All others should come at 10.00 a.m.

Q14.9.7 MCQ 2014 2M

By the beginning of the 20th century, several hypotheses were being proposed, suggesting a paradigm shift in our understanding of the universe. However, the clinching evidence was provided by experimental measurements of the position of a star which was directly behind our sun. Which of the following inference(s) may be drawn from the above passage?

(i) Our understanding of the universe changes based on the positions of stars
(ii) Paradigm shifts usually occur at the beginning of centuries
(iii) Stars are important objects in the universe
(iv) Experimental evidence was important in confirming this paradigm shift

A. (i), (ii) and (iv) **B.** (iii) only

C. (i) and (iv) **D.** (iv) only

Q14.9.8 MCQ 2014 2M

The Gross Domestic Product (GDP) in Rupees grew at 7% during 2012-2013. For international comparison, the GDP is compared in US Dollars (USD) after conversion based on the market exchange rate. During the period 2012-2013 the exchange rate for the USD increased from Rs. 50/ USD to Rs. 60/ USD. India's GDP in USD during the period 2012-2013.

A. increased by 5%

B. decreased by 20%

C. decreased by 11%

D. decreased by 13%

Q14.9.9 MCQ 2014 2M

The ratio of male to female students in a college for five years is plotted in the following line graph. If the number of female students in 2011 and 2012 is equal, what is the ratio of male students in 2012 to male students in 2011?

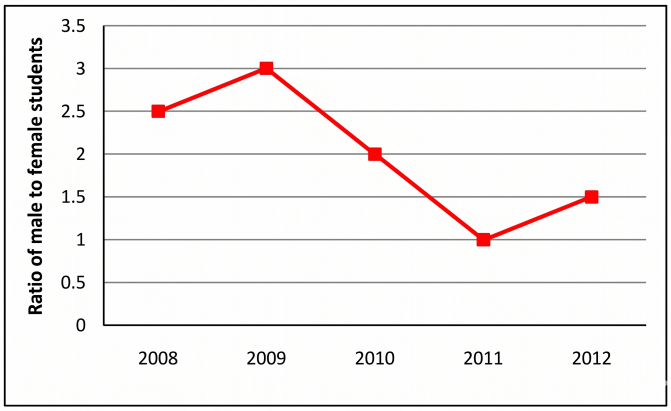


Fig. Q14.9.9

A. 1:1 **B.** 2:1 **C.** 1.5:1 **D.** 2.5:1

Q14.9.10 MCQ 2014 2M

Consider the equation: $(7526)_8 - (Y)_8 = (4364)_8$, where $(X)_N$ stands for X to the base N. Find Y.

A. 1634 **B.** 1737 **C.** 3142 **D.** 3162

Answer Key

Session : 1					
Q14.1.1 A	Q14.1.2 B	Q14.1.3 D	Q14.1.4 C	Q14.1.5 1300	Q14.1.6 D
Q14.1.7 B	Q14.1.8 180	Q14.1.9 D	Q14.1.10 B		
Session : 2					
Q14.2.1 D	Q14.2.2 A	Q14.2.3 C	Q14.2.4 25	Q14.2.5 A	Q14.2.6 A
Q14.2.7 C	Q14.2.8 D	Q14.2.9 C	Q14.2.10 0.48-0.49		
Session : 3					
Q14.3.1 C	Q14.3.2 D	Q14.3.3 B	Q14.3.4 A	Q14.3.5 725	Q14.3.6 D
Q14.3.7 B	Q14.3.8 560	Q14.3.9 D	Q14.3.10 B		
Session : 4					
Q14.4.1 B	Q14.4.2 B	Q14.4.3 C	Q14.4.4 B	Q14.4.5 45	Q14.4.6 C
Q14.4.7 D	Q14.4.8 163	Q14.4.9 D	Q14.4.10 A		
Session : 5					
Q14.5.1 C	Q14.5.2 B	Q14.5.3 B	Q14.5.4 16	Q14.5.5 D	Q14.5.6 B
Q14.5.7 D	Q14.5.8 4	Q14.5.9 20000	Q14.5.10 0.80 - 0.82		
Session : 6					
Q14.6.1 A	Q14.6.2 B	Q14.6.3 B	Q14.6.4 A	Q14.6.5 495	Q14.6.6 C
Q14.6.7 B	Q14.6.8 B	Q14.6.9 22	Q14.6.10 B		
Session : 7					
Q14.7.1 B	Q14.7.2 C	Q14.7.3 C	Q14.7.4 96	Q14.7.5 D	Q14.7.6 C
Q14.7.7 B	Q14.7.8 850	Q14.7.9 48	Q14.7.10 6		
Session : 8					
Q14.8.1 A	Q14.8.2 B	Q14.8.3 A	Q14.8.4 B	Q14.8.5 C	Q14.8.6 B
Q14.8.7 D	Q14.8.8 D	Q14.8.9 140	Q14.8.10 A		
Session : 9					
Q14.9.1 D	Q14.9.2 C	Q14.9.3 C	Q14.9.4 C	Q14.9.5 C	Q14.9.6 B
Q14.9.7 D	Q14.9.8 D	Q14.9.9 C	Q14.9.10 C		

YEAR

2013

4 Sessions · 40 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AR, CE, GG, MA, MT, PH, TF	10
Session 2	AE, AG, BT, CH, CY, MN, XE, XL	10
Session 3	CS, ME, PI	10
Session 4	EC, EE, IN	10

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 20/01/13

Q13.1.1 MCQ 2013 1M

A number is as much greater than 75 as it is smaller than 117. The number is:

A. 91 **B.** 93 **C.** 89 **D.** 96

Q13.1.2 MCQ 2013 1M

The professor (I) ordered to (II) the students to go (III) out of the class. (IV)

Which of the above underlined parts of the sentence is grammatically incorrect?

A. I **B.** II **C.** III **D.** IV

Q13.1.3 MCQ 2013 1M

Which of the following options is the closest in meaning to the word given below:

Primeval

A. Modern **B.** Historic **C.** Primitive **D.** Antique

Q13.1.4 MCQ 2013 1M

Friendship, no matter how _____ it is, has its limitations.

A. cordial **B.** intimate **C.** secret **D.** pleasant

Q13.1.5 MCQ 2013 1M

Select the pair that best expresses a relationship similar to that expressed in the pair:

Medicine: Health

A. Science: Experiment **B.** Wealth: Peace

C. Education: Knowledge **D.** Money: Happiness

Q13.1.6 MCQ 2013 2M

X and Y are two positive real numbers such that $2X+Y \leq 6$ and $X+2Y \leq 8$. For which of the following values of (X, Y) the function $f(X, Y) = 3X+6Y$ will give maximum value?

A. $(4/3, 10/3)$ **B.** $(8/3, 20/3)$

C. $(8/3, 10/3)$ **D.** $(4/3, 20/3)$

Q13.1.7 MCQ 2013 2M

If $|4X - 7| = 5$ then the values of $2|X| - |-X|$ is:

A. 2, 1/3 **B.** 1/2, 3 **C.** 3/2, 9 **D.** 2/3, 9

Q13.1.8 MCQ 2013 2M

Following table provides figures (in rupees) on annual expenditure of a firm for two years - 2010 and 2011.

Category	2010	2011
Raw material	5200	6240
Power & fuel	7000	9450
Salary & wages	9000	12600
Plant & machinery	20000	25000
Advertising	15000	19500
Research & Development	22000	26400

In 2011, which of the following two categories have registered increase by same percentage?

A. Raw material and Salary & wages

B. Salary & wages and Advertising

C. Power & fuel and Advertising

D. Raw material and Research & Development

Q13.1.9 MCQ 2013 2M

A firm is selling its product at Rs. 60 per unit. The total cost of production is Rs. 100 and firm is earning total profit of Rs. 500. Later, the total cost increased by 30%. By what percentage the price should be increased to maintained the same profit level.

A. 5 **B.** 10 **C.** 15 **D.** 30

Q13.1.10 MCQ 2013 2M

Abhishek is elder to Savar.
Savar is younger to Anshul.

Which of the given conclusions is logically valid and is inferred from the above statements?

A. Abhishek is elder to Anshul

B. Anshul is elder to Abhishek

C. Abhishek and Anshul are of the same age

D. No conclusion follows

Session : 2 | Afternoon | 20/01/13

Q13.2.1 MCQ 2013 1M

If $3 \le X \le 5$ and $8 \le Y \le 11$ then which of the following options is TRUE?

A. $\frac{3}{5} \le \frac{X}{Y} \le \frac{8}{5}$ **B.** $\frac{3}{11} \le \frac{X}{Y} \le \frac{5}{8}$

C. $\frac{3}{11} \le \frac{X}{Y} \le \frac{8}{5}$ **D.** $\frac{3}{5} \le \frac{X}{Y} \le \frac{8}{11}$

Q13.2.2 MCQ 2013 1M

The Headmaster _____ to speak to you.
Which of the following options is incorrect to complete the above sentence?

A. is wanting **B.** wants **C.** want **D.** was wanting

Q13.2.3 MCQ 2013 1M

Mahatama Gandhi was known for his humility as

A. he played an important role in humiliating exit of British from India.

B. he worked for humanitarian causes.

C. he displayed modesty in his interactions.

D. he was a fine human being.

Q13.2.4 MCQ 2024 1M

All engineering students (I) should learn mechanics, (II) mathematics and (III) how to do computation. (IV)

Which of the above underlined parts of the sentence is not appropriate?

A. I **B.** II **C.** III **D.** IV

Q13.2.5 MCQ 2013 1M

Select the pair that best expresses a relationship similar to that expressed in the pair:
water: pipe::

A. cart: road **B.** electricity: wire

C. sea: beach **D.** music: instrument

Q13.2.6 MCQ 2013 2M

Velocity of an object fired directly in upward direction is given by $V = 80 - 32t$, where t (time) is in seconds. When will the velocity be between 32 m/sec and 64 m/sec?

A. (1, 3/2) **B.** (1/2, 1) **C.** (1/2, 3/2) **D.** (1, 3)

Q13.2.7 MCQ 2013 2M

In a factory, two machines M1 and M2 manufacture 60% and 40% of the autocomponents respectively. Out of the total production, 2% of M1 and 3% of M2 are found to be defective. If a randomly drawn autocomponent from the combined lot is found defective, what is the probability that it was manufactured by M2?

A. 0.35 **B.** 0.45 **C.** 0.5 **D.** 0.4

Q13.2.8 MCQ 2013 2M

Following table gives data on tourists from different countries visiting India in the year 2011.

Country	Number of Tourists
USA	2000
England	3500
Germany	1200
Italy	1100
Japan	2400
Australia	2300
France	1000

Which two countries contributed to the one third of the total number of tourists who visited India in 2011?

A. USA and Japan **B.** USA and Australia

C. England and France **D.** Japan and Australia

Q13.2.9 MCQ 2013 2M

If $|-2X + 9| = 3$ then the possible value of $|-X| - X^2$ would be:

A. 30 **B.** -30 **C.** -42 **D.** 42

Q13.2.10 MCQ 2013 2M

All professors are researchers
Some scientists are professors

Which of the given conclusions is logically valid and is inferred from the above arguments:

A. All scientists are researchers

B. All professors are scientists

C. Some researchers are scientists

D. No conclusion follows

Session : 3 | Forenoon | 10/02/13

Q13.3.1 MCQ 2013 1M

Which one of the following options is the closest in meaning to the word given below?

Nadir

A. Highest **B.** Lowest **C.** Medium **D.** Integration

Q13.3.2 MCQ 2013 1M

Complete the sentence:

Universalism is to particularism as diffuseness is to _____.

A. specificity **B.** neutrality

C. generality **D.** adaptation

Q13.3.3 MCQ 2013 1M

What will be the maximum sum of 44, 42, 40, . . . ?

A. 502 **B.** 504 **C.** 506 **D.** 500

Q13.3.4 MCQ 2013 1M

Were you a bird, you _____ in the sky.

A. would fly **B.** shall fly

C. should fly **D.** shall have flown

Q13.3.5 MCQ 2013 1M

Choose the grammatically **INCORRECT** sentence:

A. He is of Asian origin.

B. They belonged to Africa.

C. She is an European.

D. They migrated from India to Australia.

Q13.3.6 MCQ 2013 2M

Find the sum of the expression

$$\frac{1}{\sqrt{1}+\sqrt{2}}+\frac{1}{\sqrt{2}+\sqrt{3}}+\frac{1}{\sqrt{3}+\sqrt{4}}+\cdots+\frac{1}{\sqrt{80}+\sqrt{81}}$$

A. 7 **B.** 8 **C.** 9 **D.** 10

Q13.3.7 MCQ 2013 2M

Out of all the 2-digit integers between 1 and 100, a 2-digit number has to be selected at random. What is the probability that the selected number is not divisible by 7?

A. 13/90 **B.** 12/90 **C.** 78/90 **D.** 77/90

Q13.3.8 MCQ 2013 2M

After several defeats in wars, Robert Bruce went in exile and wanted to commit suicide. Just before committing suicide, he came across a spider attempting tirelessly to have its net. Time and again, the spider failed but that did not deter it to refrain from making attempts. Such attempts by the spider made Bruce curious. Thus, Bruce started observing the near-impossible goal of the spider to have the net. Ultimately, the spider succeeded in having its net despite several failures. Such act of the spider encouraged Bruce not to commit suicide. And then, Bruce went back again and won many a battle, and the rest is history.

Which one of the following assertions is best supported by the above information?

A. Failure is the pillar of success.

B. Honesty is the best policy.

C. Life begins and ends with adventures.

D. No adversity justifies giving up hope.

Q13.3.9 MCQ 2013 2M

A tourist covers half of his journey by train at 60 km/h, half of the remainder by bus at 30 km/h and the rest by cycle at 10 km/h. The average speed of the tourist in km/h during his entire journey is

A. 36 **B.** 30 **C.** 24 **D.** 18

Q13.3.10 MCQ 2013 2M

The current erection cost of a structure is $Rs.$ 13, 200. If the labour wages per day increase by 1/5 of the current wages and the working hours decrease by 1/24 of the current period, then the new cost of erection in $Rs.$ is

A. 16,500 **B.** 15,180 **C.** 11,000 **D.** 10,120

Session : 4 | Afternoon | 10/02/13

Q13.4.1 MCQ 2013 1M

They were requested not to **quarrel** with others.
Which one of the following options is the closest in meaning to the word **quarrel**?

A. make out **B.** call out **C.** dig out **D.** fall out

Q13.4.2 MCQ 2013 1M

In the summer of 2012, in New Delhi, the mean temperature of Monday to Wednesday was 41°C and of Tuesday to Thursday was 43°C. If the temperature on Thursday was 15% higher than that of Monday, then the temperature in °C on Thursday was

A. 40 **B.** 43 **C.** 46 **D.** 49

Q13.4.3 MCQ 2013 1M

Complete the sentence:
Dare _____ mistakes.

A. commit **B.** to commit
C. committed **D.** committing

Q13.4.4 MCQ 2013 1M

Choose the grammatically **CORRECT** sentence:

A. Two and two add four.

B. Two and two become four.

C. Two and two are four.

D. Two and two make four.

Q13.4.5 MCQ 2013 1M

Statement: You can always give me a ring whenever you need.
Which one of the following is the best inference from the above statement?

A. Because I have a nice caller tune.

B. Because I have a better telephone facility.

C. Because a friend in need is a friend indeed.

D. Because you need not pay towards the telephone bills when you give me a ring.

Q13.4.6 MCQ 2013 2M

A car travels 8 km in the first quarter of an hour, 6 km in the second quarter and 16 km in the third quarter. The average speed of the car in km per hour over the entire journey is

A. 30 **B.** 36 **C.** 40 **D.** 24

Q13.4.7 MCQ 2013 2M

Find the sum to n terms of the series $10 + 84 + 734 + \ldots$

A. $\frac{9(9^n + 1)}{10} + 1$ **B.** $\frac{9(9^n - 1)}{8} + 1$

C. $\frac{9(9^n - 1)}{8} + n$ **D.** $\frac{9(9^n - 1)}{8} + n^2$

Q13.4.8 MCQ 2013 2M

Statement: There were different streams of freedom movements in colonial India carried out by the moderates, liberals, radicals, socialists, and so on.
Which one of the following is the best inference from the above statement?

A. The emergence of nationalism in colonial India led to our Independence.

B. Nationalism in India emerged in the context of colonialism.

C. Nationalism in India is homogeneous.

D. Nationalism in India is heterogeneous.

Q13.4.9 MCQ 2013 2M

The set of values of p for which the roots of the equation $3x^2 + 2x + p(p - 1) = 0$ are of opposite sign is

A. $(-\infty, 0)$ **B.** $(0, 1)$ **C.** $(1, \infty)$ **D.** $(0, \infty)$

Q13.4.10 MCQ 2013 2M

What is the chance that a leap year, selected at random, will contain 53 Saturdays?

A. 2/7 **B.** 3/7 **C.** 1/7 **D.** 5/7

Answer Key

Session : 1					
Q13.1.1 D	Q13.1.2 B	Q13.1.3 C	Q13.1.4 B	Q13.1.5 C	Q13.1.6 A
Q13.1.7 B	Q13.1.8 D	Q13.1.9 A	Q13.1.10 D		
Session : 2					
Q13.2.1 B	Q13.2.2 C	Q13.2.3 C	Q13.2.4 D	Q13.2.5 B	Q13.2.6 C
Q13.2.7 C	Q13.2.8 C	Q13.2.9 B	Q13.2.10 C		
Session : 3					
Q13.3.1 B	Q13.3.2 A	Q13.3.3 C	Q13.3.4 A	Q13.3.5 C	Q13.3.6 B
Q13.3.7 D	Q13.3.8 D	Q13.3.9 C	Q13.3.10 B		
Session : 4					
Q13.4.1 D	Q13.4.2 C	Q13.4.3 B	Q13.4.4 D	Q13.4.5 C	Q13.4.6 C
Q13.4.7 D	Q13.4.8 D	Q13.4.9 B	Q13.4.10 A		

PrepFusion

YEAR

2012

4 Sessions · 40 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	AR, GG, TF	10
Session 2	AE, AG, MN	10
Session 3	BT, CE, CH, CS, ME, PH, PI	10
Session 4	CY, EC, EE, IN, MA, MT, XE, XL	10

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 29/01/12

Q12.1.1 MCQ 2012 1M

Which one of the following options is the closest in meaning to the word given below?

Pacify

A. Excite **B.** Soothe **C.** Deplete **D.** Tire

Q12.1.2 MCQ 2012 1M

Choose the most appropriate pair of words from the options given below to complete the following sentence:

The high level of _____ of the questions in the test was _____ by an increase in the period of time allotted for answering them.

A. difficulty, compensated **B.** exactitude, magnified

C. aptitude, decreased **D.** attitude, mitigated

Q12.1.3 MCQ 2012 1M

Choose the grammatically **CORRECT** sentence:

A. He laid in bed till 8 o'clock in the morning.

B. He layed in bed till 8 o'clock in the morning.

C. He lain in bed till 8 o'clock in the morning.

D. He lay in bed till 8 o'clock in the morning.

Q12.1.4 MCQ 2012 1M

Which one of the parts (A, B, C, D) in the sentence contains an **ERROR**?

No sooner had the doctor seen the results of the blood test, than he suggested the patient to see the specialist.

A. no sooner had **B.** results of the blood test

C. suggested the patient **D.** see the specialist

Q12.1.5 MCQ 2012 1M

Ten teams participate in a tournament. Every team plays each of the other teams twice. The total number of matches to be played is

A. 20 **B.** 45 **C.** 60 **D.** 90

Q12.1.6 MCQ 2012 2M

A value of x that satisfies the equation $log\ x + log\ (x - 7) = log\ (x + 11) + log\ 2$ is

A. 1 **B.** 2 **C.** 7 **D.** 11

Q12.1.7 MCQ 2012 2M

Let $f(x) = x - [x]$, where $x \geq 0$ and $[x]$ is the greatest integer not larger than x. Then $f(x)$ is a

A. monotonically increasing function

B. monotonically decreasing function

C. linearly increasing function between two integers

D. linearly decreasing function between two integers

Q12.1.8 MCQ 2012 2M

Ravi is taller than Arun but shorter than Iqbal. Sam is shorter than Ravi. Mohan is shorter than Arun. Balu is taller than Mohan and Sam. The tallest person can be

A. Mohan **B.** Ravi **C.** Balu **D.** Arun

Q12.1.9 MCQ 2012 2M

A smuggler has 10 capsules in which five are filled with narcotic drugs and the rest contain the original medicine. All the 10 capsules are mixed in a single box, from which the customs officials picked two capsules at random and tested for the presence of narcotic drugs. The probability that the smuggler will be caught is

A. 0.50 **B.** 0.67 **C.** 0.78 **D.** 0.82

Q12.1.10 MCQ 2012 2M

The documents expose the cynicism of the government officials – and yet as the media website reflects, not a single newspaper has reported on their existence.

Which one of the following inferences may be drawn with the greatest accuracy from the above passage?

A. Nobody other than the government officials knew about the existence of the documents.

B. Newspapers did report about the documents but nobody cared.

C. Media reports did not show the existence of the documents.

D. The documents reveal the attitude of the government officials.

Session : 2 | Afternoon | 29/01/12

Q12.2.1 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
I _____ to have bought a diamond ring.

A. have a liking **B.** should have liked

C. would like **D.** may like

Q12.2.2 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
Food prices _____ again this month.

A. have raised **B.** have been raising

C. have been rising **D.** have arose

Q12.2.3 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
The administrators went on to implement yet another unreasonable measure, arguing that the measures were already _____ and one more would hardly make a difference.

A. reflective **B.** utopian **C.** luxuriant **D.** unpopular

Q12.2.4 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:
To those of us who had always thought him timid, his ____ came as a surprise.

A. intrepidity **B.** inevitability **C.** inability **D.** inertness

Q12.2.5 MCQ 2012 1M

The arithmetic mean of five different natural numbers is 12. The largest possible value among the numbers is

A. 12 **B.** 40 **C.** 50 **D.** 60

Q12.2.6 MCQ 2012 2M

Two policemen, A and B, fire once each at the same time at an escaping convict. The probability that A hits the convict is three times the probability that B hits the convict. If the probability of the convict not getting injured is 0.5, the probability that B hits the convict is

A. 0.14 **B.** 0.22 **C.** 0.33 **D.** 0.40

Q12.2.7 MCQ 2012 2M

The total runs scored by four cricketers P, Q, R, and S in years 2009 and 2010 are given in the following table:

Player	2009	2010
P	802	1008
Q	765	912
R	429	619
S	501	701

The player with the lowest percentage increase in total runs is

A. P **B.** Q **C.** R **D.** S

Q12.2.8 MCQ 2012 2M

If a prime number on division by 4 gives a remainder of 1, then that number can be expressed as

A. sum of squares of two natural numbers

B. sum of cubes of two natural numbers

C. sum of square roots of two natural numbers

D. sum of cube roots of two natural numbers

Q12.2.9 MCQ 2012 2M

Two points $(4, p)$ and $(0, q)$ lie on a straight line having a slope of $3/4$. The value of $(p - q)$ is

A. -3 **B.** 0 **C.** 3 **D.** 4

Q12.2.10 MCQ 2012 2M

In the early nineteenth century, theories of social evolution were inspired less by Biology than by the conviction of social scientists that there was a growing improvement in social institutions. Progress was taken for granted and social scientists attempted to discover its laws and phases.

Which one of the following inferences may be drawn with the greatest accuracy from the above passage?

Social scientists

A. did not question that progress was a fact.

B. did not approve of Biology.

C. framed the laws of progress.

D. emphasized Biology over Social Sciences.

Session : 3 | Forenoon | 12/02/12

Q12.3.1 MCQ 2012 1M

The cost function for a product in a firm is given by $5q^2$, where q is the amount of production. The firm can sell the product at a market price of ₹50 per unit. The number of units to be produced by the firm such that the profit is maximized is

A. 5 **B.** 10 **C.** 15 **D.** 25

Q12.3.2 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:

Despite several _____ the mission succeeded in its attempt to resolve the conflict.

A. attempts **B.** setbacks

C. meetings **D.** delegations

Q12.3.3 MCQ 2012 1M

Which one of the following options is the closest in meaning to the word given below?

Mitigate

A. Diminish **B.** Divulge **C.** Dedicate **D.** Denote

Q12.3.4 MCQ 2012 1M

Choose the grammatically **INCORRECT** sentence:

A. They gave us the money back less the service charges of Three Hundred rupees.

B. This country's expenditure is not less than that of Bangladesh.

C. The committee initially asked for a funding of Fifty Lakh rupees, but later settled for a lesser sum.

D. This country's expenditure on educational reforms is very less.

Q12.3.5 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:

Suresh's dog is the one _____ was hurt in the stampede.

A. that **B.** which **C.** who **D.** whom

Q12.3.6 MCQ 2012 2M

Wanted Temporary, Part-time persons for the post of Field Interviewer to conduct personal interviews to collect and collate economic data. Requirements: High School-pass, must be available for Day, Evening and Saturday work. Transportation paid, expenses reimbursed.

Which one of the following is the best inference from the above advertisement?

A. Gender-discriminatory

B. Xenophobic

C. Not designed to make the post attractive

D. Not gender-discriminatory

Q12.3.7 MCQ 2012 2M

A political party orders an arch for the entrance to the ground in which the annual convention is being held. The profile of the arch follows the equation $y = 2x - 0.1x^2$ where y is the height of the arch in meters. The maximum possible height of the arch is

A. 8 meters **B.** 10 meters **C.** 12 meters **D.** 14 meters

Q12.3.8 MCQ 2012 2M

An automobile plant contracted to buy shock absorbers from two suppliers X and Y. X supplies 60% and Y supplies 40% of the shock absorbers. All shock absorbers are subjected to a quality test. The ones that pass the quality test are considered reliable. Of X's shock absorbers, 96% are reliable. Of Y's shock absorbers, 72% are reliable.

The probability that a randomly chosen shock absorber, which is found to be reliable, is made by Y is

A. 0.288 **B.** 0.334 **C.** 0.667 **D.** 0.720

Q12.3.9 MCQ 2012 2M

Which of the following assertions are **CORRECT**?

P: Adding 7 to each entry in a list adds 7 to the mean of the list

Q: Adding 7 to each entry in a list adds 7 to the standard deviation of the list

R: Doubling each entry in a list doubles the mean of the list

S: Doubling each entry in a list leaves the standard deviation of the list unchanged

A. P, Q **B.** Q, R **C.** P, R **D.** R, S

Q12.3.10 MCQ 2012 2M

Given the sequence of terms, AD, CG, FK, JP, the next term is

A. OV **B.** OW **C.** PV **D.** PW

Session : 4 | Afternoon | 12/02/12

Q12.4.1 MCQ 2012 1M

If $(1.001)^{1259} = 3.52$ and $(1.001)^{2062} = 7.85$, then $(1.001)^{3321} =$

A. 2.23 **B.** 4.33 **C.** 11.37 **D.** 27.64

Q12.4.2 MCQ 2012 1M

Choose the most appropriate alternative from the options given below to complete the following sentence:

If the tired soldier wanted to lie down, he _____ the mattress out on the balcony.

A. should take **B.** shall take

C. should have taken **D.** will have taken

Q12.4.3 MCQ 2012 1M

Choose the most appropriate word from the options given below to complete the following sentence:

Given the seriousness of the situation that he had to face, his _____ was impressive.

A. beggary **B.** nomenclature

C. jealousy **D.** nonchalance

Q12.4.4 MCQ 2012 1M

Which one of the following options is the closest in meaning to the word given below?

Latitude

A. Eligibility **B.** Freedom

C. Coercion **D.** Meticulousness

Q12.4.5 MCQ 2012 1M

One of the parts (A, B, C, D) in the sentence given below contains an ERROR. Which one of the following is **INCORRECT**?

I requested that he should be given the driving test today instead of tomorrow.

A. requested that **B.** should be given

C. the driving test **D.** instead of tomorrow

Q12.4.6 MCQ 2012 2M

One of the legacies of the Roman legions was discipline. In the legions, military law prevailed and discipline was brutal. Discipline on the battlefield kept units obedient, intact and fighting, even when the odds and conditions were against them.

Which one of the following statements best sums up the meaning of the above passage?

A. Thorough regimentation was the main reason for the efficiency of the Roman legions even in adverse circumstances.

B. The legions were treated inhumanly as if the men were animals.

C. Discipline was the armies' inheritance from their seniors.

D. The harsh discipline to which the legions were subjected to led to the odds and conditions being against them.

Q12.4.7 MCQ 2012 2M

Raju has 14 currency notes in his pocket consisting of only Rs. 20 notes and Rs. 10 notes. The total money value of the notes is Rs. 230. The number of Rs. 10 notes that Raju has is

A. 5 **B.** 6 **C.** 9 **D.** 10

Q12.4.8 MCQ 2012 2M

There are eight bags of rice looking alike, seven of which have equal weight and one is slightly heavier. The weighing balance is of unlimited capacity. Using this balance, the minimum number of weighings required to identify the heavier bag is

A. 2 **B.** 3 **C.** 4 **D.** 8

Q12.4.9 MCQ 2012 2M

The data given in the following table summarizes the monthly budget of an average household.

Category	Amount (Rs.)
Food	4000
Clothing	1200
Rent	2000
Savings	1500
Other expenses	1800

The approximate percentage of the monthly budget **NOT** spent on savings is

A. 10% **B.** 14% **C.** 81% **D.** 86%

Q12.4.10 MCQ 2012 2M

A and B are friends. They decide to meet between 1 PM and 2 PM on a given day. There is a condition that whoever arrives first will not wait for the other for more than 15 minutes. The probability that they will meet on that day is

A. 1/4 **B.** 1/16 **C.** 7/16 **D.** 9/16

PrepFusion

Answer Key

Session : 1					
Q12.1.1 B	Q12.1.2 A	Q12.1.3 D	Q12.1.4 C	Q12.1.5 D	Q12.1.6 D
Q12.1.7 C	Q12.1.8 C	Q12.1.9 C	Q12.1.10 D		
Session : 2					
Q12.2.1 C	Q12.2.2 C	Q12.2.3 D	Q12.2.4 A	Q12.2.5 C	Q12.2.6 A
Q12.2.7 B	Q12.2.8 A	Q12.2.9 C	Q12.2.10 A		
Session : 3					
Q12.3.1 A	Q12.3.2 B	Q12.3.3 A	Q12.3.4 D	Q12.3.5 Marks to All	Q12.3.6 D
Q12.3.7 B	Q12.3.8 B	Q12.3.9 C	Q12.3.10 A		
Session : 4					
Q12.4.1 D	Q12.4.2 A	Q12.4.3 D	Q12.4.4 B	Q12.4.5 B	Q12.4.6 A
Q12.4.7 A	Q12.4.8 A	Q12.4.9 D	Q12.4.10 C		

PrepFusion

YEAR

2011

4 Sessions · 40 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	GG, TF	10
Session 2	AE, MN	10
Session 3	AR, BT, CE, CH, CS, ME, PH, PI	10
Session 4	AG, CY, EC, EE, IN, MA, MT, XE, XL	10

Year Snapshot

Total Questions: 40 **MCQ:** 40 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 30/01/11

Q11.1.1 MCQ 2011 1M

Choose the most appropriate word or phrase from the options given below to complete the following sentence.

The environmentalists hope _____ the lake to its pristine condition.

A. in restoring **B.** in the restoration of

C. to restore **D.** restoring

Q11.1.2 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Polemical

A. imitative **B.** conciliatory **C.** truthful **D.** ideological

Q11.1.3 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

Despite the mixture's _____ nature, we found that by lowering its temperature in the laboratory we could dramatically reduce its tendency to vaporize.

A. acerbic **B.** resilient **C.** volatile **D.** heterogeneous

Q11.1.4 MCQ 2011 1M

If m students require a total of m pages of stationery in m days, then 100 students will require 100 pages of stationery in

A. 100 days **B.** $m/100$ days **C.** $100/m$ days **D.** m days

Q11.1.5 MCQ 2011 1M

Choose the most appropriate words from the options given below to complete the following sentence:

Because she had a reputation for _____, we were surprised and pleased when she greeted us so _____.

A. insolence . . . . . irately

B. insouciance . . . . . curtly

C. graciousness . . . . . amiably

D. querulousness . . . . . affably

Q11.1.6 MCQ 2011 2M

The number of solutions for the following system of inequalities is

$$X_1 \geq 0$$
$$X_2 \geq 0$$
$$X_1 + X_2 \leq 10$$
$$2X_1 + 2X_2 \geq 22$$

A. 0 **B.** infinite **C.** 1 **D.** 2

Q11.1.7 MCQ 2011 2M

In a class of 300 students in an M.Tech programme, each student is required to take at least one subject from the following three:

M600: Advanced Engineering Mathematics
C600: Computational Methods for Engineers
E600: Experimental Techniques for Engineers

The registration data for the M.Tech class shows that 100 students have taken M600, 200 students have taken C600, and 60 students have taken E600. What is the maximum possible number of students in the class who have taken all the above three subjects?

A. 20 **B.** 30 **C.** 40 **D.** 50

Q11.1.8 MCQ 2011 2M

Three sisters (R, S, and T) received a total of 24 toys during Christmas. The toys were initially divided among them in a certain proportion. Subsequently, R gave some toys to S which doubled the share of S. Then S in turn gave some of her toys to T, which doubled T's share. Next, some of T's toys were given to R, which doubled the number of toys that R currently had. As a result of all such exchanges, the three sisters were left with equal number of toys. How many toys did R have originally?

A. 8 **B.** 9 **C.** 11 **D.** 12

Q11.1.9 MCQ 2011 2M

The quality of services delivered by a company consists of six factors as shown below in the radar diagram. The dots in the figure indicate the score for each factor on a scale of 0 to 10. The standardized coefficient for each factor is given in the parentheses. The contribution of each factor to the overall service quality is directly proportional to the factor score and its standardized coefficient.

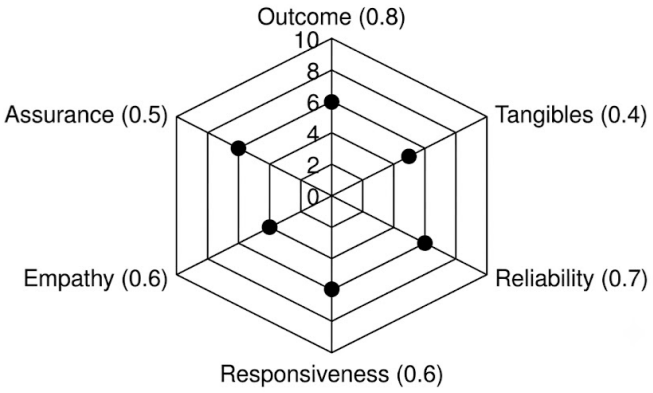


Fig. Q11.1.9

The lowest contribution among all the above factors to the overall quality of services delivered by the company is

A. 10% **B.** 20% **C.** 24% **D.** 40%

Q11.1.10 MCQ 2011 2M

In order to develop to full potential, a baby needs to be physically able to respond to the environment.

It can be inferred from the passage that

A. Full physical potential is needed in order for a baby to be able to respond to the environment.

B. It is necessary for a baby to be able to physically respond to the environment for it to develop its full potential.

C. Response to the environment of physically able babies needs to be developed to its full potential.

D. A physically able baby needs to develop its full potential in order to respond to its environment.

Session : 2 | Afternoon | 30/01/11

Q11.2.1 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Deference

A. aversion **B.** resignation

C. suspicion **D.** contempt

Q11.2.2 MCQ 2011 1M

Choose the most appropriate word(s) from the options given below to complete the following sentence:

We lost confidence in him because he never _____ the grandiose promises he had made.

A. delivered **B.** delivered on **C.** forgot **D.** reneged on

Q11.2.3 MCQ 2011 1M

Choose the word or phrase that best completes the sentence below.

_____ in the frozen wastes of Arctic takes special equipment.

A. To survive **B.** Surviving

C. Survival **D.** That survival

Q11.2.4 MCQ 2011 1M

In how many ways 3 scholarships can be awarded to 4 applicants, when each applicant can receive any number of scholarships?

A. 4 **B.** 12 **C.** 64 **D.** 81

Q11.2.5 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

The _____ of evidence was on the side of the plaintiff since all but one witness testified that his story was correct.

A. paucity **B.** propensity

C. preponderance **D.** accuracy

Q11.2.6 MCQ 2011 2M

If $(2y + 1)/(y + 2) < 1$, then which of the following alternatives gives the **CORRECT** range of y?

A. $-2 < y < 2$ **B.** $-2 < y < 1$

C. $-3 < y < 1$ **D.** $-4 < y < 1$

Q11.2.7 MCQ 2011 2M

A student attempted to solve a quadratic equation in x twice. However, in the first attempt, he incorrectly wrote the constant term and ended up with the roots as $(4, 3)$. In the second attempt, he incorrectly wrote down the coefficient of x and got the roots as $(3, 2)$. Based on the above information, the roots of the correct quadratic equation are

A. (-3, 4) **B.** (3, -4) **C.** (6, 1) **D.** (4, 2)

Q11.2.8 MCQ 2011 2M

L, M and N are waiting in a queue meant for children to enter the zoo. There are 5 children between L and M, and 8 children between M and N. If there are 3 children ahead of N and 21 children behind L, then what is the minimum number of children in the queue?

A. 28 **B.** 27 **C.** 41 **D.** 40

Q11.2.9 MCQ 2011 2M

Four archers P, Q, R and S try to hit a bull's eye during a tournament consisting of seven rounds. As illustrated in the figure below, a player receives 10 points for hitting the bulls' eye, 5 points for hitting within the inner circle and 1 point for hitting within the outer circle.

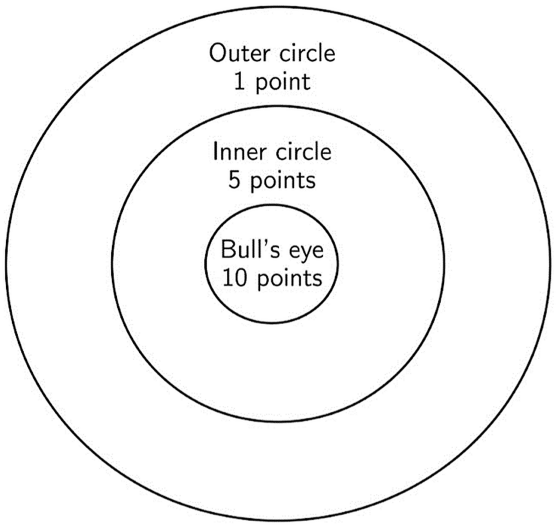


Fig. Q11.2.9

The final scores received by the players during the tournament are listed in the table below.

Round	P	Q	R	S
1	1	5	1	10
2	5	10	10	1
3	1	1	1	5
4	10	10	1	1
5	1	5	5	10
6	10	5	1	1
7	5	10	1	1

The most accurate and the most consistent players during the tournament are respectively

A. P and S **B.** Q and R **C.** Q and Q **D.** R and Q

Q11.2.10 MCQ 2013 2M

Nimbus clouds are dark and ragged, stratus clouds appear dull in colour and cover the entire sky. Cirrus clouds are thin and delicate, whereas cumulus clouds look like cotton balls.

It can be inferred from the passage that

A. A cumulus cloud on the ground is called fog

B. It is easy to predict the weather by studying clouds

C. Clouds are generally of very different shapes, sizes and mass

D. There are four basic cloud types: stratus, nimbus, cumulus and cirrus

Session : 3 | Forenoon | 13/02/11

Q11.3.1 MCQ 2011 1M

Choose the most appropriate word(s) from the options given below to complete the following sentence.

I contemplated _____ Singapore for my vacation but decided against it.

A. to visit **B.** having to visit

C. visiting **D.** for a visit

Q11.3.2 MCQ 2011 1M

If $\log(P) = (1/2)\log(Q) = (1/3)\log(R)$, then which of the following options is **TRUE**?

A. $P^2 = Q^3R^2$ **B.** $Q^2 = PR$

C. $Q^2 = R^3P$ **D.** $R = P^2Q^2$

Q11.3.3 MCQ 2011 1M

Which of the following options is the closest in the meaning to the word below:

Inexplicable

A. Incomprehensible **B.** Indelible

C. Inextricable **D.** Infallible

Q11.3.4 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Amalgamate

A. merge **B.** split **C.** collect **D.** separate

Q11.3.5 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

If you are trying to make a strong impression on your audience, you cannot do so by being understated, tentative or _____.

A. hyperbolic **B.** restrained

C. argumentative **D.** indifferent

Q11.3.6 MCQ 2011 2M

The variable cost (V) of manufacturing a product varies according to the equation $V = 4q$, where q is the quantity produced. The fixed cost (F) of production of same product reduces with q according to the equation $F = 100/q$. How many units should be produced to minimize the total cost ($V + F$)?

A. 5 **B.** 4 **C.** 7 **D.** 6

Q11.3.7 MCQ 2011 2M

P, Q, R and S are four types of dangerous microbes recently found in a human habitat. The area of each circle with its diameter printed in brackets represents the growth of a single microbe surviving human immunity system within 24 hours of entering the body. The danger to human beings varies proportionately with the toxicity, potency and growth attributed to a microbe shown in the figure below:

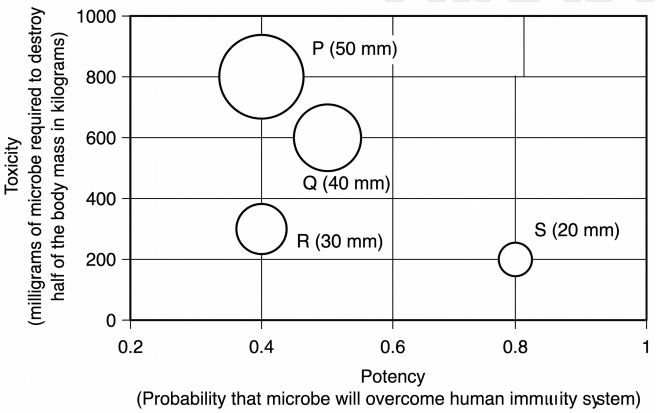


Fig. Q11.3.7

A pharmaceutical company is contemplating the development of a vaccine against the most dangerous microbe. Which microbe should the company target in its first attempt?

A. P **B.** Q **C.** R **D.** S

Q11.3.8 MCQ 2011 2M

Few school curricula include a unit on how to deal with bereavement and grief, and yet all students at some point in their lives suffer from losses through death and parting.

Based on the above passage which topic would not be included in a unit on bereavement?

A. how to write a letter of condolence

B. what emotional stages are passed through in the healing process

C. what the leading causes of death are

D. how to give support to a grieving friend

Q11.3.9 MCQ 2011 2M

A container originally contains 10 litres of pure spirit. From this container 1 litre of spirit is replaced with 1 litre of water. Subsequently, 1 litre of the mixture is again replaced with 1 litre of water and this process is repeated one more time. How much spirit is now left in the container?

A. 7.58 litres **B.** 7.84 litres **C.** 7 litres **D.** 7.29 litres

Q11.3.10 MCQ 2011 2M

A transporter receives the same number of orders each day. Currently, he has some pending orders (backlog) to be shipped. If he uses 7 trucks, then at the end of the 4th day he can clear all the orders. Alternatively, if he uses only 3 trucks, then all the orders are cleared at the end of the 10th day. What is the minimum number of trucks required so that there will be no pending order at the end of the 5th day?

A. 4 **B.** 5 **C.** 6 **D.** 7

Session : 4 | Afternoon | 13/02/11

Q11.4.1 MCQ 2011 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair:

Gladiator : Arena

A. dancer : stage **B.** commuter : train

C. teacher : classroom **D.** lawyer : courtroom

Q11.4.2 MCQ 2011 1M

There are two candidates P and Q in an election. During the campaign, 40% of the voters promised to vote for P, and the rest for Q. However, on the day of election 15% of the voters went back on their promise to vote for P and instead voted for Q. 25% of the voters went back on their promise to vote for Q and instead voted for P. Suppose, P lost by 2 votes, then what was the total number of voters?

A. 100 **B.** 110 **C.** 90 **D.** 95

Q11.4.3 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

It was her view that the country's problems had been _____ by foreign technocrats, so that to invite them to come back would be counter-productive.

A. identified **B.** ascertained

C. exacerbated **D.** analysed

Q11.4.4 MCQ 2011 1M

Choose the word from the options given below that is most nearly opposite in meaning to the given word:

Frequency

A. periodicity **B.** rarity **C.** gradualness **D.** persistency

Q11.4.5 MCQ 2011 1M

Choose the most appropriate word from the options given below to complete the following sentence:

Under ethical guidelines recently adopted by the Indian Medical Association, human genes are to be manipulated only to correct diseases for which _____ treatments are unsatisfactory.

A. similar **B.** most **C.** uncommon **D.** available

Q11.4.6 MCQ 2011 2M

The horse has played a little known but very important role in the field of medicine. Horses were injected with toxins of diseases until their blood built up immunities. Then a serum was made from their blood. Serums to fight with diphtheria and tetanus were developed this way.

It can be inferred from the passage, that horses were

A. given immunity to diseases

B. generally quite immune to diseases

C. given medicines to fight toxins

D. given diphtheria and tetanus serums

Q11.4.7 MCQ 2011 2M

The fuel consumed by a motorcycle during a journey while traveling at various speeds is indicated in the graph below.

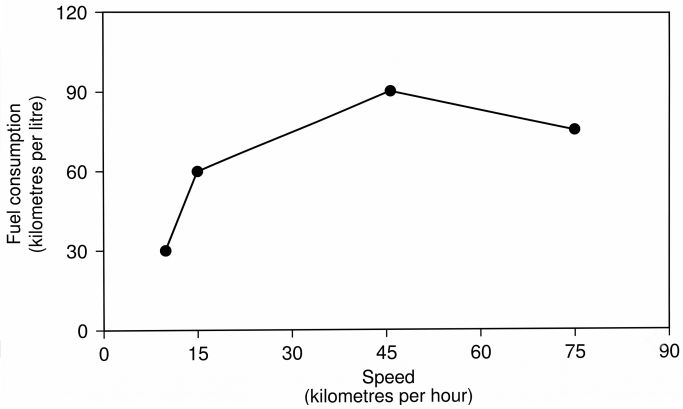


Fig. Q11.4.7

The distances covered during four laps of the journey are listed in the table below

Lap	Distance (kilometres)	Average speed (kilometres per hour)
P	15	15
Q	75	45
R	40	75
S	10	10

From the given data, we can conclude that the fuel consumed per kilometre was least during the lap

A. P **B.** Q **C.** R **D.** S

Q11.4.8 MCQ 2011 2M

Three friends, R, S and T shared toffee from a bowl. R took $1/3^{rd}$ of the toffees, but returned four to the bowl. S took $1/4^{th}$ of what was left but returned three toffees to the bowl. T took half of the remainder but returned two back into the bowl. If the bowl had 17 toffees left, how many toffees were originally there in the bowl?

A. 38 **B.** 31 **C.** 48 **D.** 41

Q11.4.9 MCQ 2011 2M

Given that $f(y) = |y|/y$, and q is any non-zero real number, the value of $|f(q) - f(-q)|$ is

A. 0 **B.** -1 **C.** 1 **D.** 2

Q11.4.10 MCQ 2011 2M

The sum of n terms of the series $4 + 44 + 444 + \ldots$ is

A. $(4/81)[10^{n+1} - 9n - 1]$

B. $(4/81)[10^{n-1} - 9n - 1]$

C. $(4/81)[10^{n+1} - 9n - 10]$

D. $(4/81)[10^{n} - 9n - 10]$

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Answer Key

Session : 1					
Q11.1.1 C	Q11.1.2 B	Q11.1.3 C	Q11.1.4 D	Q11.1.5 D	Q11.1.6 A
Q11.1.7 B	Q11.1.8 C	Q11.1.9 A	Q11.1.10 B		
Session : 2					
Q11.2.1 D	Q11.2.2 B	Q11.2.3 A	Q11.2.4 C	Q11.2.5 C	Q11.2.6 B
Q11.2.7 C	Q11.2.8 A	Q11.2.9 B	Q11.2.10 D		
Session : 3					
Q11.3.1 C	Q11.3.2 B	Q11.3.3 A	Q11.3.4 B	Q11.3.5 B	Q11.3.6 A
Q11.3.7 D	Q11.3.8 C	Q11.3.9 D	Q11.3.10 C		
Session : 4					
Q11.4.1 D	Q11.4.2 A	Q11.4.3 C	Q11.4.4 B	Q11.4.5 D	Q11.4.6 A
Q11.4.7 B	Q11.4.8 C	Q11.4.9 D	Q11.4.10 C		

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YEAR

2010

3 Sessions · 30 Questions

Sessions Summary

Session	Participating Branches	Total Qs
Session 1	TF	10
Session 2	MN	10
Session 3	EC, EE, IN, CS, ME, CE	10

Year Snapshot

Total Questions: 30 **MCQ:** 30 **MSQ:** 0 **NAT:** 0

Session : 1 | Forenoon | 07/02/10

Q10.1.1 MCQ 2010 1M

Which of the following options is the closest in meaning to the word below:
Ephemeral

A. effeminate **B.** ghostlike **C.** soft **D.** short-lived

Q10.1.2 MCQ 2010 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.
Erudition : Scholar

A. steadfast : mercurial **B.** competence : strict

C. skill : craftsman **D.** nurse : doctor

Q10.1.3 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:
The two child norm with _____ for the violators will have significant implications for our demographic profile.

A. disincentives **B.** incitements

C. restrictions **D.** restraints

Q10.1.4 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:
There is no fixed relation between food and famine; famines can occur with or without substantial _____ in food output.

A. aberration **B.** weakening

C. decline **D.** deterioration

Q10.1.5 MCQ 2010 1M

Consider the function $f(x) = \max(7 - x, x + 3)$. In which range does f take its minimum value?

A. $-6 \le x < -2$ **B.** $-2 \le x < 2$

C. $2 \le x < 6$ **D.** $6 \le x < 10$

Q10.1.6 MCQ 2010 2M

It has taken fifty six long and frustrating years to turn bronze into gold for India's Olympics aspirations. Beijing 2008 marks a defining moment in India's Olympic history. From Delhi to Beijing is a long journey but one that our Olympians have undertaken with courage.

Which of the following statements best sums up the meaning of the above passage?

A. India's participation in Olympics has been frustrating.

B. Beijing Olympics was a landmark in India's Olympic history.

C. Our Olympians have undertaken a long journey to Beijing.

D. India's bronze medal turned into gold at Beijing.

Q10.1.7 MCQ 2010 2M

Consider the series

$$\frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \frac{1}{8} + \frac{1}{9} - \frac{1}{16} + \frac{1}{32} + \frac{1}{27} - \frac{1}{64} + \ldots.$$

The sum of the infinite series above is:

A. ∞ **B.** $\frac{5}{6}$ **C.** $\frac{1}{2}$ **D.** 0

Q10.1.8 MCQ 2010 2M

A gathering of 50 linguists discovered that 4 knew Kannada, Telugu and Tamil, 7 knew only Telugu and Tamil, 5 knew only Kannada and Tamil, 6 knew only Telugu and Kannada. If the number of linguists who knew Tamil is 24 and those who knew Kannada is also 24, how many linguists knew only Telugu?

A. 9 **B.** 10 **C.** 11 **D.** 8

Q10.1.9 MCQ 2010 2M

A tank has 100 litres of water. At the end of every hour the following two operations are performed in sequence: i) water equal to $m\%$ of the current contents of the tank is added to the tank, ii) water equal to $n\%$ of the current contents of the tank is removed from the tank. At the end of 5 hours the tank contains exactly 100 litres of water. The relation between m and n is:

A. m=n **B.** m>n

C. m<n **D.** None of the previous

Q10.1.10 MCQ 2010 2M

A student is answering a multiple choice examination with 65 questions with a marking scheme as follows: i) 1 mark for each correct answer, ii) $-\frac{1}{4}$ for a wrong answer, iii) $-\frac{1}{8}$ for a question that has not been attempted. If the student gets 37 marks in the test then the least possible number of questions the student has **NOT** answered is:

A. 6 **B.** 5 **C.** 7 **D.** 4

Session : 2 | Afternoon | 07/02/10

Q10.2.1 MCQ 2010 1M

Which of the following options is the closest in meaning to the word below:

Exhort

A. urge **B.** condemn **C.** restrain **D.** scold

Q10.2.2 MCQ 2010 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.

Preamble : Constitution

A. amendment : law **B.** prologue : play

C. episode : serial **D.** plot : story

Q10.2.3 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:

The committee wrote a _____ report, extolling only the strengths of the proposal.

A. reasonable **B.** supportive **C.** biased **D.** fragmented

Q10.2.4 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:

If the country has to achieve real prosperity, it is _____ that the fruits of progress reach all, and in equal measure.

A. inevitable **B.** contingent **C.** oblivious **D.** imperative

Q10.2.5 MCQ 2010 1M

A person invests Rs. 1000 at 10% annual compound interest for 2 years. At the end of two years the whole amount is invested at an annual simple interest of 12% for 5 years. The total value of the investment finally is:

A. 1776 **B.** 1760 **C.** 1920 **D.** 1936

Q10.2.6 MCQ 2010 2M

The ban on smoking in designated public places can save a large number of people from the well known effects of environmental tobacco smoke. Passive smoking seriously impairs respiratory health. The ban rightly seeks to protect non-smokers from its ill effects.

Which of the following statements best sums up the meaning of the above passage?

A. Effects of environmental tobacco are well known.

B. The ban on smoking in public places protects the non smokers.

C. Passive smoking is bad for health.

D. The ban on smoking in public places excludes passive smoking.

Q10.2.7 MCQ 2010 2M

Given the sequence A, B, B, C, C, C, D, D, D, D, . . . etc., that is one A, two Bs, three Cs, four Ds, five Es and so on, the 240th letter in the sequence will be:

A. V **B.** U **C.** T **D.** W

Q10.2.8 MCQ 2010 2M

Consider the set of integers $\{1, 2, 3, \ldots, 5000\}$. The number of integers that is divisible by neither 3 nor 4 is:

A. 1668 **B.** 2084 **C.** 2500 **D.** 2916

Q10.2.9 MCQ 2010 2M

A positive integer m in base 10 when represented in base 2 has the representation p and in base 3 has the representation q. We get $p - q = 990$ where the subtraction is done in base 10. Which of the following is necessarily true:

A. $m \geq 14$ **B.** $9 \leq m \leq 13$ **C.** $6 \leq m \leq 8$ **D.** $m < 6$

Q10.2.10 MCQ 2010 2M

Given the following four functions $f_1(n) = n^{100}$, $f_2(n) = (1.2)^n$, $f_3(n) = 2^{n/2}$, $f_4(n) = 3^{n/3}$ which function will have the largest value for sufficiently large values of n (i.e. $n \to \infty$)?

A. f_4 **B.** f_3 **C.** f_2 **D.** f_1

Session : 3 | Forenoon | 14/02/10

Q10.3.1 MCQ 2010 1M

Which of the following options is the closest in meaning to the word below:
Circuitous

A. cyclic **B.** indirect **C.** confusing **D.** crooked

Q10.3.2 MCQ 2010 1M

The question below consists of a pair of related words followed by four pairs of words. Select the pair that best expresses the relation in the original pair.
Unemployed : Worker

A. fallow : land **B.** unaware : sleeper

C. wit : jester **D.** renovated : house

Q10.3.3 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:
If we manage to _____ our natural resources, we would leave a better planet for our children.

A. uphold **B.** restrain **C.** cherish **D.** conserve

Q10.3.4 MCQ 2010 1M

Choose the most appropriate word from the options given below to complete the following sentence:

His rather casual remarks on politics _____ his lack of seriousness about the subject.

A. masked **B.** belied **C.** betrayed **D.** suppressed

Q10.3.5 MCQ 2010 1M

25 persons are in a room. 15 of them play hockey. 17 of them play football and 10 of them play both hockey and football. Then the number of persons playing neither hockey nor football is:

A. 2 **B.** 17 **C.** 13 **D.** 3

Q10.3.6 MCQ 2010 2M

Modern warfare has changed from large scale clashes of armies to suppression of civilian populations. Chemical agents that do their work silently appear to be suited to such warfare; and regretfully, there exist people in military establishments who think that chemical agents are useful tools for their cause.

Which of the following statements best sums up the meaning of the above passage?

A. Modern warfare has resulted in civil strife.

B. Chemical agents are useful in modern warfare.

C. Use of chemical agents in warfare would be undesirable.

D. People in military establishments like to use chemical agents in war.

Q10.3.7 MCQ 2010 2M

If 137 + 276 = 435 how much is 731 + 672?

A. 534 **B.** 1403 **C.** 1623 **D.** 1513

Q10.3.8 MCQ 2010 2M

5 skilled workers can build a wall in 20 days; 8 semi-skilled workers can build a wall in 25 days; 10 unskilled workers can build a wall in 30 days. If a team has 2 skilled, 6 semi-skilled and 5 unskilled workers, how long will it take to build the wall?

A. 20 days **B.** 18 days **C.** 16 days **D.** 15 days

Q10.3.9 MCQ 2010 2M

Given digits 2, 2, 3, 3, 3, 4, 4, 4, 4 how many distinct 4 digit numbers greater than 3000 can be formed?

A. 50 **B.** 51 **C.** 52 **D.** 54

Q10.3.10 MCQ 2010 2M

Hari (H), Gita (G), Irfan (I) and Saira (S) are siblings (i.e. brothers and sisters). All were born on 1st January. The age difference between any two successive siblings (that is born one after another) is less than 3 years. Given the following facts:

i. Hari's age + Gita's age > Irfan's age + Saira's age.
ii. The age difference between Gita and Saira is 1 year. However, Gita is not the oldest and Saira is not the youngest.
iii. There are no twins.

In what order were they born (oldest first)?

A. HSIG **B.** SGHI **C.** IGSH **D.** IHSG

Answer Key

Session : 1					
Q10.1.1 D	Q10.1.2 C	Q10.1.3 A	Q10.1.4 C	Q10.1.5 C	Q10.1.6 B
Q10.1.7 B	Q10.1.8 C	Q10.1.9 B	Q10.1.10 A		
Session : 2					
Q10.2.1 A	Q10.2.2 B	Q10.2.3 C	Q10.2.4 D	Q10.2.5 D	Q10.2.6 B
Q10.2.7 A	Q10.2.8 C	Q10.2.9 B	Q10.2.10 A		
Session : 3					
Q10.3.1 B	Q10.3.2 A	Q10.3.3 D	Q10.3.4 C	Q10.3.5 D	Q10.3.6 C
Q10.3.7 C	Q10.3.8 D	Q10.3.9 B	Q10.3.10 C		

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